

Does it transfer? A pluripotency-domain benchmark of temporal trajectory-inference methods across iPSC reprogramming systems, with scANVI-validated cell-state references

Kuma^{*1}

¹Graduate School of Frontier Sciences, The University of Tokyo

Working draft, June 2026

Abstract

scTimeBench (Osakwe et al., 2026) established that temporal single-cell trajectory-inference methods should be judged on three tasks — forecast accuracy, embedding coherence, and lineage fidelity — and showed, across heterogeneous developmental datasets, that strong forecasters need not preserve biology and that lineage fidelity is the hardest task. We narrow the benchmark to a single biological domain — **induced pluripotency** — and make the reprogramming/differentiation modality the experimental axis, assembling time-labeled iPSC scRNA-seq datasets that reach or leave the pluripotent state by distinct routes, to ask whether a method’s ranking *transfers* across them. Because most iPSC datasets lack per-cell author labels, we also contribute a reference-construction-and-validation pipeline: silver cell-state references are built by scANVI/scArches label transfer and screened by a geometric out-of-distribution (OOD) gate, computed inside the main pipeline, that quantifies when a transferred reference can be trusted. Three findings stand out. **(1) Transferred references fail detectably across modality.** A cross-modality transfer collapses — the geometric OOD gate flags 73–76% of query cells while scANVI’s classifier confidence flags $\leq 0.3\%$ — so references are accepted only for within-modality transfers (agreement 0.76–0.86; labeled self-transfer 0.94), the regime the benchmark itself recommends. **(2) Lineage rankings do not transfer across systems.** Recovery against the frozen coarse reference graphs reorders the methods from one system to the next; across the four marker-silver systems lineage rankings are near-discordant (Kendall’s $W = 0.19$), with the top and bottom projection-capable methods swapping places between a mouse and a human-blood system, whereas forecast rankings stay comparatively stable ($W = 0.75$). **(3) Single-seed leaderboards are unstable.** Across five seeds the lineage leader on the primary datasets changes, and single-seed values frequently fall outside the 5-seed 95% confidence interval; a diffusion-pseudotime axis can sharpen recovery (most strikingly PRESCIENT on GSE230659) but the gain is method- and system-dependent. The practical message is that in this domain a method should be chosen per task and validated on a system close to the target biology — a lineage result in particular should not be read off a single averaged leaderboard. The central message is methodological: because rankings move with the task, the reference construction, and the random seed, a domain benchmark should report task-stratified results, an out-of-distribution check on reference transfer, and seed uncertainty rather than a single leaderboard. All datasets are public; reported numbers are read from synced result files, with multi-seed intervals provided as derived artifacts (Section 5).

^{*}Corresponding author. Co-authors and affiliations to be added.

1 Introduction

Temporal single-cell methods project cells from one observed time point to the next, and benchmarking them is hard because each cell is measured once, so the “truth” for a projected cell must be assembled from held-out time points and reference annotations. scTimeBench (Osakwe et al., 2026) formalized this with three tasks — forecast accuracy, embedding coherence, and lineage fidelity — across eight developmental datasets spanning four species, concluding that strong forecasters often fail to preserve latent-space and lineage structure, and that lineage recovery is the central weakness, with methods frequently no better than a Spearman-correlation baseline.

scTimeBench answers “which method is best on average across developmental systems?”. We ask two questions it does not. **(1) Within one domain, does method performance transfer across biological routes?** Induced pluripotency is reached or left by mechanistically distinct paths — chemical reprogramming, OSKM transcription-factor reprogramming, directed differentiation, and organoid development — that share the pluripotent state. If a method’s ranking is stable across these routes, a practitioner can choose once; if it inverts, method choice must be conditioned on the system. **(2) How do we benchmark on the many iPSC datasets that lack author labels?** Embedding coherence and lineage fidelity both need a cell-state reference, yet two of our primary datasets ship none. We address this with a scANVI/scArches label-transfer pipeline gated by a geometric out-of-distribution score, which lets the benchmark scale to unlabeled data while measuring when a transferred reference can be trusted. These are the two innovations that distinguish this work from scTimeBench. Concretely, four features set this benchmark apart from prior trajectory-inference comparisons: a single-domain (pluripotency) focus that makes cross-system ranking *transfer* testable in the first place, scTimeBench-style task stratification, an out-of-distribution-gated reference-transfer step, and an explicit, quantified treatment of single-seed ranking instability.

2 Methods

2.1 Datasets: the iPSC suite and its run status

The benchmark spans six time-labeled iPSC scRNA-seq datasets across reprogramming and differentiation modalities, plus a human cerebral-organoid time course held out as a planned extension (Table 1). Run status is heterogeneous and stated per-dataset rather than aggregated: four datasets carry completed formal runs across all three tasks (the two human chemical primaries and the mouse and human-blood cross-system datasets), GSE242424 (OSKM) carries all three tasks under a separate author-cluster reference protocol (Section 2.5), GSE175634 carries forecast and embedding only, and the organoid course is integration-ready but not yet run. All are public; provenance and per-task run status are in `datasets/dataset_inventory.{md,csv}`.

2.2 Tasks and metrics (inherited from scTimeBench)

Forecast Accuracy uses the four `geomloss` metrics (Wasserstein, Gaussian MMD, Energy-MMD, Hausdorff), recomputed centrally (`metric_backend=scTimeBench_exact`) from each method’s standardized `projected_expression.npy`. Embedding Coherence reports the ARI of Leiden clusters on the projected embedding against reference labels plus the average normalized classifier entropy. Lineage Fidelity binarizes the predicted state-transition matrix at the precision/recall-maximizing

Table 1. The pluripotency-domain dataset suite (six iPSC datasets plus an organoid extension). “Run status” is the honest state of the synced results.

Accession	System	Modality	Cells	Run status
GSE178325	human hADSC→hCiPS	chemical reprog.	80,475	results (all 3 tasks)
GSE230659	human hADSC→hCiPS	chemical reprog.	75,194	results (all 3 tasks)
GSE242424	human fibroblast→iPSC	OSKM reprog.	59,187	results (all 3 tasks)
GSE175634	human iPSC→cardiomyocyte	differentiation	224,202	forecast + embedding
GSE298212	human blood→hCiPS	chemical reprog.	38,517	results (all 3 tasks)
GSE218855	mouse MEF→iPSC	chemical reprog.	54,380	results (all 3 tasks)
E-MTAB-7552	human cerebral organoid	organoid (extension)	~43,498	integration-ready

threshold and scores single-step and multi-step (Floyd–Warshall reachability) AUROC, AUPRC, and Jaccard against the reference graph.

2.3 Methods and capability gating

Following scTimeBench’s rule that methods are scored only on supported dimensions (Table 2), the two optimal-transport methods are lineage-only; the projection-capable methods are scored on all three tasks. CellRank2 (Weiler et al., 2024) runs as fate/lineage post-processing on the WOT (Schiebinger et al., 2019) `RealTimeKernel` and is reported as such. Five projection-capable or lineage-only methods carry completed runs and constitute the evaluated roster (Table 2). Three further generative models from the scTimeBench roster — scIMF (Jiang et al., 2026) (a Transformer + interacting mean-field neural SDE), PI-SDE (Jiang and Wan, 2024) (a physics-informed neural SDE), and Squidiff (He et al., 2026) (a conditional diffusion model) — are scoped as **planned extensions**: their adapter interfaces, runtime configs, and Shirokane job are in the repository, but the model code is not yet vendored (the adapters raise an explicit `VendoringRequired` contract), so these methods are *not* evaluated here and are excluded from every results table and figure. They are listed separately below to document the intended capability gating, not as integrated results.

Table 2. Method roster and scTimeBench capability gating. The upper block is the evaluated roster (completed runs); the lower block lists planned extensions whose model code is not yet vendored and which appear in no results table or figure.

Method	Algorithm	Projection-capable	Dimensions scored
<i>Evaluated roster (completed runs)</i>			
WOT	Optimal transport	No	Lineage only
CellRank2	Fate mapping on WOT <code>RealTimeKernel</code>	No	Lineage only
scNODE	VAE + neural ODE	Yes	All three
PRESCIENT	PCA + neural SDE	Yes	All three
MIOFlow	Geodesic VAE + neural ODE	Yes	All three
<i>Planned extensions (adapter contract only — not evaluated here)</i>			
scIMF	Transformer + interacting mean-field SDE	Yes	(planned: all three)
PI-SDE	Physics-informed neural SDE	Yes	(planned: all three)
Squidiff	Conditional diffusion (PCA space)	Yes	(planned: all three)

2.4 Scenarios

For each dataset, Scenario A is observed-time interpolation, Scenario B is extrapolation to later unseen time points, and Scenario C is the joint setting; metrics are computed on held-out time points. Scenarios D/E/F repeat the same interpolation/extrapolation/joint splits on a 15-bin diffusion-pseudotime axis (Scanpy DPT, rooted at the earliest observed time point), testing whether ordering cells by an inferred internal clock improves recovery.

2.5 Reference construction and validation (the scANVI integration)

Because the chemical-reprogramming primaries ship no per-cell author labels, silver references are built by scANVI/scArches (Xu et al., 2021; Lotfollahi et al., 2022) label transfer from an annotated atlas and screened by three gates: a **self-recovery gate** (scANVI must re-predict the reference’s own labels at ≥ 0.90 ; met on GSE242424 at 0.990), a **geometric OOD gate** (each query cell’s mean distance to its k nearest reference cells in the shared latent, referenced against the reference’s own kNN-distance distribution; a cell is flagged when it falls beyond the reference’s q -quantile, with defaults $k = 15$, $q = 0.95$), and a **within-modality rule**. The geometric gate is computed inside the main pipeline (`src/pipeline.py` calls `src/annotation/ood.py`) and reported alongside the classifier-confidence flag, so each run emits both the global OOD fraction and a per-state breakdown; the same routine runs on the server scANVI latent via `scripts/score_ood.py`, which is now a thin wrapper over the shared gate. The threshold parameters (k, q) are configurable; a sensitivity sweep (`scripts/ood_sensitivity_sweep.py`, on the local PCA-latent proxy) shows the cross-modality reject decision is not an artifact of these choices — the chemical query is flagged $6.5\text{--}9.7\times$ more often than the reference’s own self-flag baseline across every $k \in \{5, 15, 30, 50\}$ and $q \in \{0.90, 0.95\}$, collapsing toward parity only in the extreme $q = 0.99$ tail where the proxy cannot resolve the boundary. Accepted transfers seed the frozen `official_silver` provider; rejected (cross-modality) cases fall back to marker-defined references. The unit-tested utilities are in `src/annotation/`.

3 Results

3.1 No method dominates all three tasks

On the two chemical-reprogramming datasets, the combined embedding-plus-lineage ranking over the projection-capable methods is narrow — MIOFlow first (combined rank score 1.79), PRESCIENT second (1.85), scNODE third (2.18) — but the per-task picture is split (Table 3): PRESCIENT has the most coherent embeddings (mean ARI 0.135), MIOFlow recovers lineage best (mean single-step AUROC 0.662), and forecast accuracy has no single winner (scNODE wins Scenario A on both datasets; PRESCIENT wins the harder B/C on GSE230659). The best forecaster is not the best lineage recoverer — the central scTimeBench observation, reproduced on reprogramming data. The complete results matrix — all five completed methods on all six systems, with forecast, embedding, and lineage side by side — is given in Table 5; the main-text tables and figures are task- and role-stratified views of it (projection-capable methods only for forecast/embedding, OSKM kept on its own reference protocol).

Table 3. Primary chemical-reprogramming benchmark (GSE178325 + GSE230659, Scenarios A/B/C), projection-capable methods. Means over datasets and scenarios; higher ARI and AUROC are better.

Method	Combined rank score	Mean embedding ARI	Mean single-step AUROC	Best at
MIOFlow	1.79	0.072	0.662	Lineage fidelity
PRESCIENT	1.85	0.135	0.630	Embedding coherence
scNODE	2.18	0.054	0.597	Forecast (Scenario A)

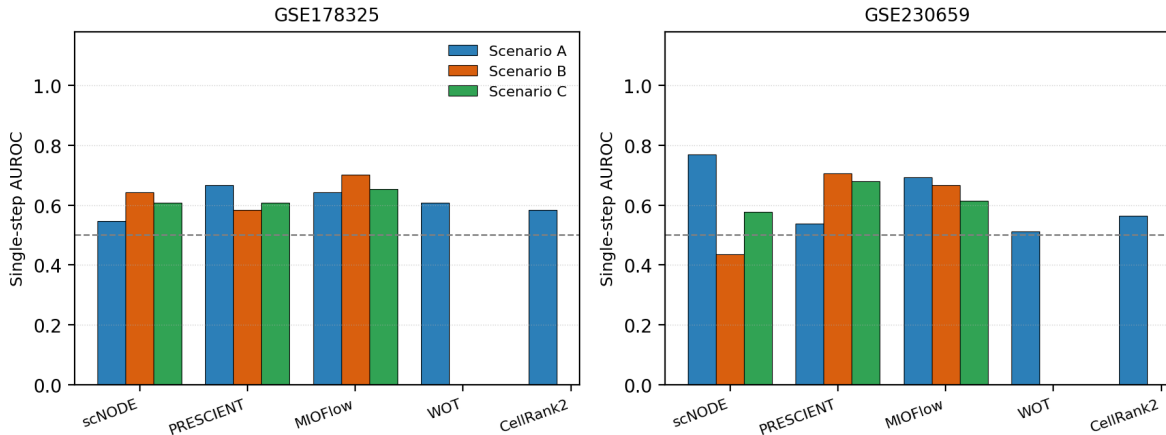


Figure 1. Lineage Fidelity (graph-sim single-step AUROC) by method and scenario for the two chemical datasets. WOT/CellRank2 are lineage-only (run for Scenario A). Absolute AUROC stays in a modest 0.44–0.77 band; the dashed line is chance.

3.2 Forecasting difficulty: extrapolation degrades every method

Scenario B is the universal hard case (Fig. 2): on GSE230659 the forecast Wasserstein distance rises from ≈ 81 (scNODE, A) to 148–404 (B), and the lowest lineage AUROC in the primary benchmark — scNODE on GSE230659 B at 0.436 — falls below chance. Early-only training leaves the late reprogramming milestones unobserved, and extrapolation is exposed to library-specific shifts.

3.3 Cross-modality generalization: rankings do not transfer

The headline result of the pluripotency-domain design is that the *same* methods behave very differently across reprogramming routes (Fig. 3). Lineage single-step AUROC on the OSKM system (GSE242424) reaches 0.85 (Scenario A) with method means of 0.71–0.78, versus 0.59–0.66 on chemical GSE230659 — and the method ordering is not preserved. Lineage performance is therefore as much a property of the biological system and its reference-graph structure as of the method: a single averaged leaderboard would hide this, and a method validated on one reprogramming route cannot be assumed to rank the same on another. Figure 3 shows the same instability across all five systems; the next two subsections extend it beyond reprogramming route to a different species and starting cell type, and put a concordance statistic on it.

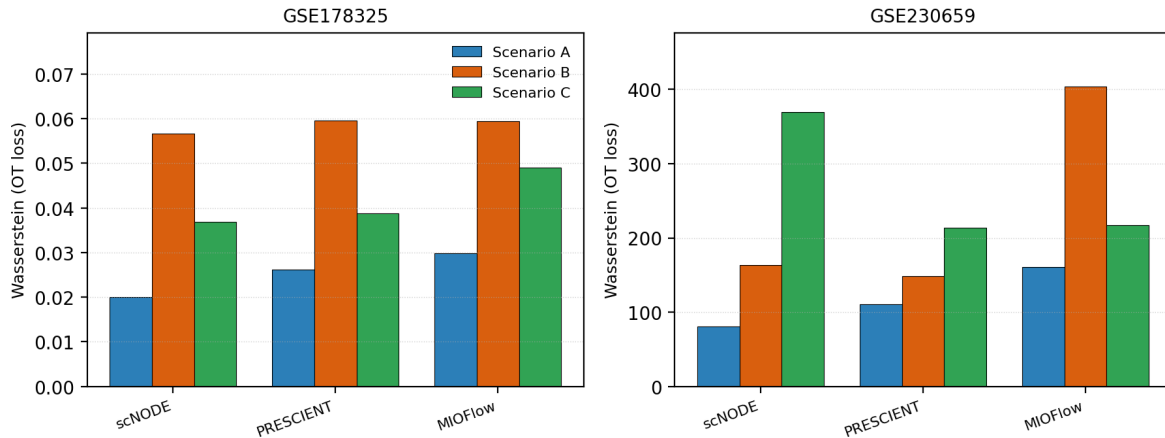


Figure 2. Forecast Accuracy (exact OT-loss Wasserstein, lower is better) by method and scenario; the extrapolation setting inflates the loss for all methods. Note the different vertical scales.

3.4 Cross-system replication: the ranking flip generalizes to mouse and blood

Extending the suite beyond a single species and starting cell type sharpens the same conclusion (Table 4). On the mouse MEF→iPSC system (GSE218855), scNODE is the strongest projection-capable lineage method (mean single-step AUROC 0.87 over A/B/C) and PRESCIENT the weakest (0.78); on the human blood→hCiPS system (GSE298212) the order *inverts* — PRESCIENT leads (0.86) and scNODE trails (0.74). The embedding-coherence ordering flips too (PRESCIENT best on mouse at ARI 0.23, MIOFlow best on blood at 0.17), while forecast error stays tightly bunched on both (Wasserstein 0.075–0.076 on mouse, 0.044–0.054 on blood). The lineage-only optimal-transport baselines remain competitive at the trajectory-root scenario (CellRank2 single-step AUROC 0.92 on mouse, 0.82 on blood). That the top and bottom generative methods swap places between a mouse and a human system — neither of which shares the hADSC starting state used to develop the methods — is direct evidence that a leaderboard from one system does not transfer, now replicated across species and across starting cell type rather than reprogramming route alone.

Table 4. Cross-system replication on the two added datasets. Mean over Scenarios A/B/C for the three projection-capable methods (single-step lineage AUROC, exact OT-loss forecast Wasserstein [lower better], embedding-coherence ARI). The lineage ordering inverts between the mouse and human-blood systems.

Dataset	Method	Lineage AUROC	Forecast WD	Embed. ARI
GSE218855 (mouse)	scNODE	0.87	0.075	0.14
	PRESCIENT	0.78	0.075	0.23
	MIOFlow	0.76	0.076	0.19
GSE298212 (blood)	scNODE	0.74	0.045	0.06
	PRESCIENT	0.86	0.044	0.09
	MIOFlow	0.80	0.054	0.17

3.5 Ranking (in)stability is task-dependent

To put a statistic on the transfer question we rank the three projection-capable methods within each of the four marker-silver datasets (mean over Scenarios A/B/C) and compute Kendall’s coefficient

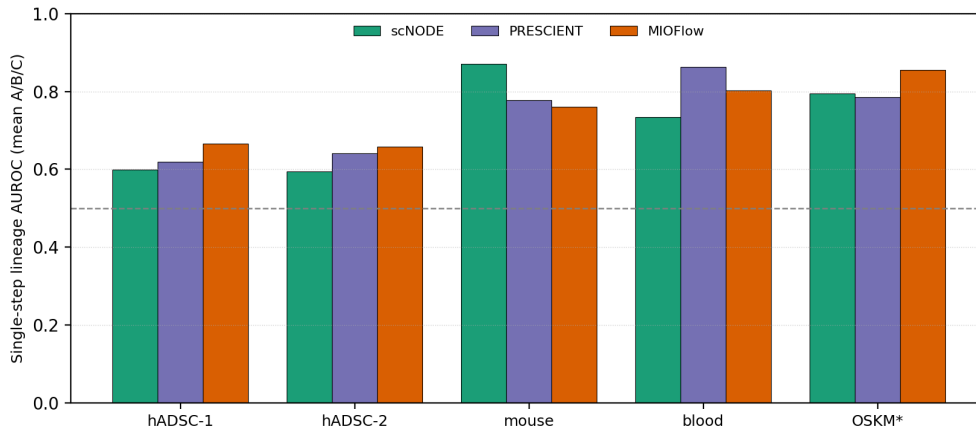


Figure 3. Cross-system generalization. Lineage single-step AUROC (mean over A/B/C) for the three projection-capable methods across the four marker-silver systems (hADSC-1 = GSE178325, hADSC-2 = GSE230659, mouse = GSE218855, blood = GSE298212) and the OSKM system (GSE242424; *evaluated under the author-cluster reference protocol, so its AUROC is not strictly comparable to the marker-silver bars). The leading method changes from MIOFlow (hADSC) to scNODE (mouse) to PRESCIENT (blood); absolute AUROC is also markedly higher away from the two hADSC systems.

of concordance W across datasets, separately for each task (Fig. 4). The three tasks behave very differently. Lineage fidelity is the *least* stable: $W = 0.19$, near the no-agreement floor — the two human hADSC→hCiPS systems rank the methods identically (pairwise Spearman $\rho = +1.0$), but the mouse system inverts that order completely ($\rho = -1.0$ against both), and the human-blood system sits in between. Forecast accuracy is the *most* stable ($W = 0.75$: scNODE/PRESCIENT tend to lead the OT-loss everywhere), with embedding coherence intermediate ($W = 0.56$). A single averaged lineage leaderboard is therefore actively misleading, whereas a forecast leaderboard generalizes better — a distinction a one-number summary would erase. The same scope-sensitivity appears in the combined embedding-plus-lineage rank: MIOFlow leads on the observed-time primary benchmark (Table 3), but PRESCIENT takes first once the pseudotime scenarios or the mouse/blood systems enter the pool. Which method “wins” is a function of the evaluation scope, not a fixed property.

3.6 Robustness: seed sensitivity and a correlation floor

Two sanity checks qualify the point estimates above. First, repeating the primary runs across five random seeds (same hyperparameters; Fig. 5) shows the single-seed leaderboard is not robust. Averaged over seeds, the best lineage recoverer on the two chemical primaries is *PRESCIENT* (mean single-step AUROC 0.78), not MIOFlow (0.75) or scNODE (0.71) — the reverse of the single-seed ordering in Table 3, and the single-seed values (open circles) frequently fall outside the 5-seed 95% confidence interval. The seed sensitivity is itself method-specific: PRESCIENT’s lineage output is essentially seed-invariant (per-scenario SD ≤ 0.008), whereas MIOFlow’s forecast Wasserstein on GSE230659 swings wildly (Scenario B mean 529, 95% CI ± 203). That even the random seed can flip the lineage leader is the strongest form of the instability this benchmark documents, and it means single-run leaderboards in this domain should be read as draws, not rankings.

Second, the scTimeBench Spearman correlation baseline (adjacent-timepoint, maximum-vote) sets a floor every method should clear. It is low on the two human hADSC systems (single-step AUROC

Method-ranking (in)stability across biological systems

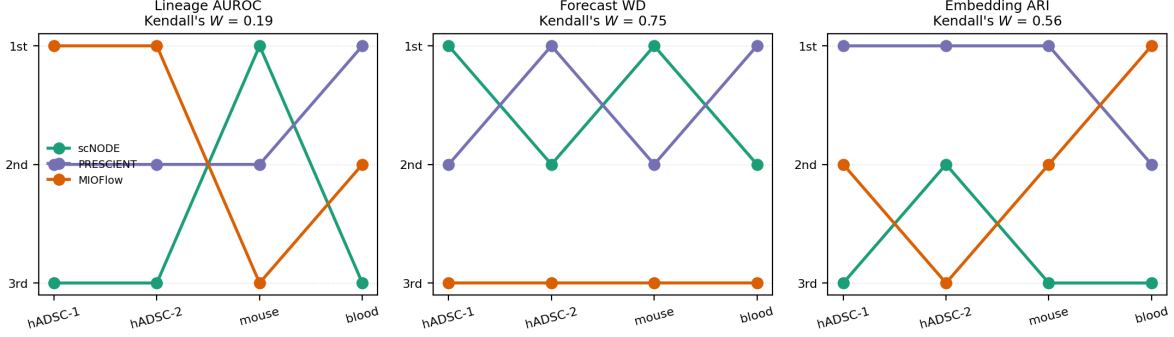


Figure 4. Method-ranking (in)stability across the four marker-silver systems (hADSC-1 = GSE178325, hADSC-2 = GSE230659, mouse = GSE218855, blood = GSE298212). Each line tracks one method’s within-dataset rank (mean over A/B/C); Kendall’s W is annotated per task. Lineage rankings cross repeatedly ($W = 0.19$); forecast rankings are nearly flat ($W = 0.75$).

0.45 and 0.50), which all methods beat comfortably, but high on the cross-system datasets (0.72 mouse, 0.79 blood) — where the methods only marginally exceed it (mouse means 0.76–0.87; blood 0.74–0.86). Much of the apparently strong off-domain lineage recovery is thus attributable to simple adjacent-timepoint expression similarity rather than learned dynamics, reinforcing scTimeBench’s caution that lineage fidelity is the field’s weakest task.

Multi-seed robustness (bars = mean $\pm$ 95% CI over 5 seeds; open circles = single-seed value)

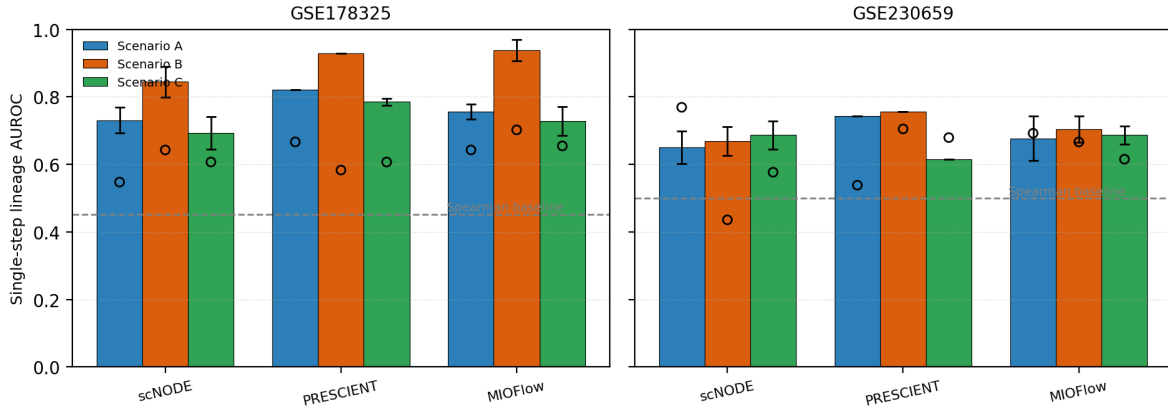


Figure 5. Multi-seed robustness of single-step lineage AUROC on the two primary datasets. Bars are the mean over five seeds with 95% confidence intervals; open circles are the single-seed (seed 42) values reported in the main tables; the dashed line is the scTimeBench Spearman correlation baseline. Single-seed points often lie outside the 5-seed interval, and the seed-averaged leader (PRESCIENT) differs from the single-seed leader (MIOFlow).

3.7 A label-permutation negative control bounds the metric’s chance level

To show the lineage metric does not return a plausible score regardless of input, we recomputed the single-step graph-similarity AUROC (the production metric) on every real predicted transition

matrix — the three projection methods across all six systems and every available scenario (observed-time A/B/C and pseudotime D/E/F; 72 runs) — under two null inputs: a *label-permutation* null that randomly relabels the predicted states (2000 draws) and a *random row-stochastic matrix* control (2000 draws); Figure 6. On the small reference graphs the empirical chance level is AUROC ≈ 0.62 (permutation) and ≈ 0.63 (random) — above the nominal 0.5, because a 4–5-node chain and a diagonal-heavy row-stochastic prediction are easy to match partially. Against this empirical null only 23 of 72 runs exceed the null’s 95th percentile. The OSKM system (GSE242424) is the clearest signal (above null at $p \leq 0.012$ across A/B/C), whereas the human hADSC silver systems (GSE178325, GSE230659) on observed time frequently are not — scNODE on GSE230659 B (AUROC 0.44) sits *below* its permutation null ($p = 0.91$). The pseudotime axis is method-specific: PRESCIENT’s much-cited jump on GSE230659 is real — it clears the null in every pseudotime scenario (D/E/F AUROC 0.95–1.00, $p \leq 0.045$) — but most other pseudotime gains do not exceed chance (only 4 of 18 pseudotime runs clear the null). The metric is therefore discriminating rather than permissive, and absolute lineage AUROC on the silver systems should be read against an empirical permutation floor near 0.62, not against 0.5 — reinforcing the central caution that these silver-reference lineage scores are weak signal except where a permutation test confirms otherwise.

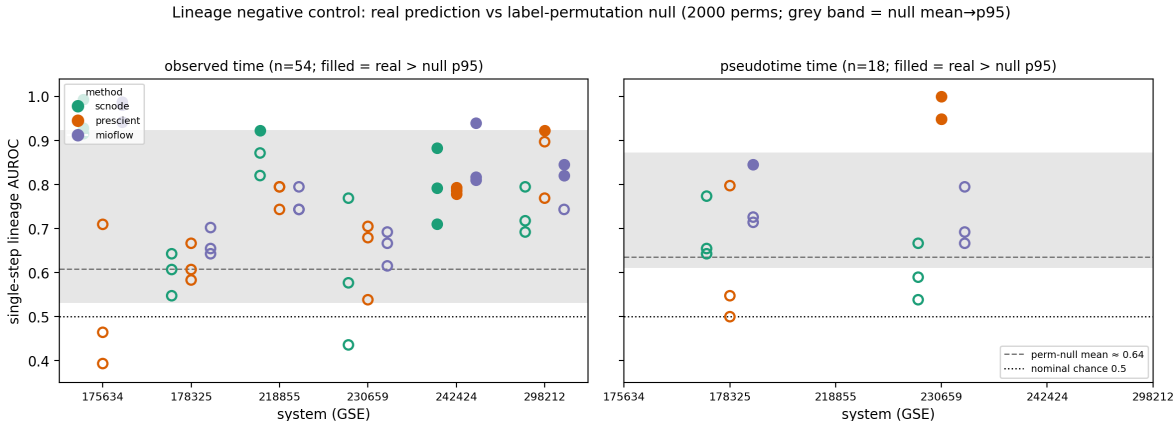


Figure 6. Lineage negative control across all 72 method $\times$ system $\times$ scenario runs (observed A/B/C and pseudotime D/E/F). Each point is the real single-step AUROC (filled = above the label-permutation null’s 95th percentile; open = at/below it); grey marks the permutation-null mean (2000 permutations). The empirical chance level sits near 0.62 on these small graphs; real scores clear it only where genuine lineage signal exists (23/72 runs), including PRESCIENT’s pseudotime jump on GSE230659 but few other pseudotime gains.

3.8 The scANVI reference gate flags cross-modality transfers that confidence misses

The reference-construction pipeline (Section 2.5) is validated against a positive and a negative control (Fig. 7). The **positive control** is a *cross-modality* transfer — mapping the chemical queries onto the OSKM reference, which biology says should be rejected: it collapses, with 76–96% of query cells confidently assigned to a single OSKM-transgene state ($< 1\%$ of the reference) and agreement with the existing labels of only 0.01–0.04, yet scANVI’s posterior confidence flags only $\leq 0.3\%$ of cells. The geometric OOD gate flags **75.9%/73.0%** of query cells on the server scANVI latent, correctly catching the mismatch confidence misses, and the flagged fraction is concentrated in the biologically plausible states rather than spread uniformly (per-state breakdown

in results/.../ood_by_group_summary.json). The **negative controls** show the gate is not simply pessimistic: a *within-modality* (chemical→chemical) transfer agrees with the existing annotation at **0.76–0.86**, and a labeled self-transfer recovers held-out labels at **0.944**. The reject decision is stable to the gate’s k and quantile parameters over a sensible range (Section 2.5). The benchmark therefore accepts scANVI references only for within-modality transfers that pass the gate — exactly the regime in which the cross-modality generalization result (Fig. 3) says references should not be shared. We describe this as a screened, controlled gate rather than a fully calibrated classifier: the controls bound its behaviour at the two extremes, but a continuous calibration curve across intermediate transfer distances is left to future work.

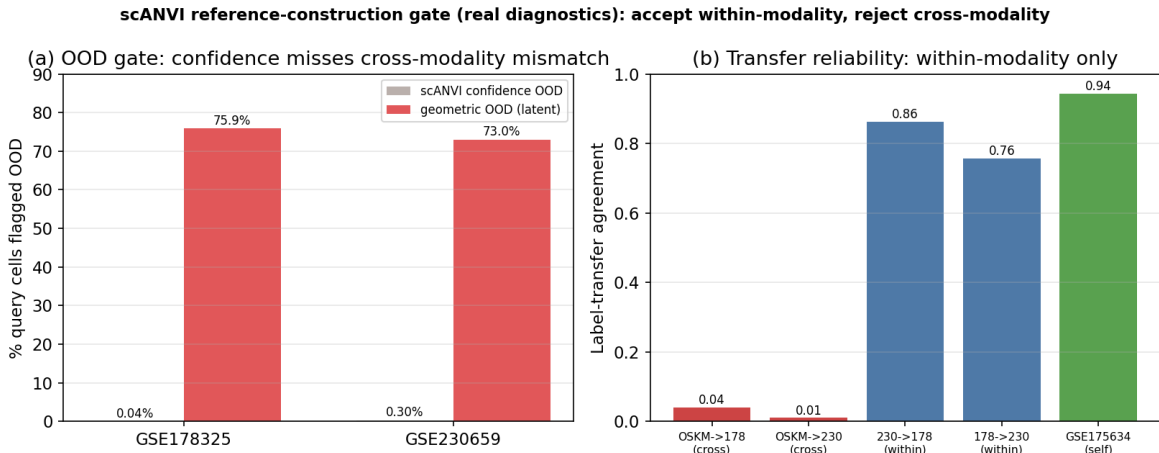


Figure 7. scANVI reference-construction gate (real diagnostics). (a) Classifier confidence flags $\leq 0.3\%$ of query cells as out-of-distribution while the geometric latent OOD score flags 73–76% — confidence is blind to the cross-modality mismatch. (b) Label-transfer agreement is near zero for cross-modality transfers but 0.76–0.86 within modality and 0.94 for a labeled self-transfer; references are accepted only in the latter regimes.

3.9 Pseudotime denoises lineage recovery, conditionally

Replacing the observed-time axis with a diffusion-pseudotime axis (Scenarios D/E/F) improves lineage fidelity, but the benefit is method- and system-dependent (Fig. 8). The effect is dramatic for PRESCIENT on GSE230659, whose mean single-step lineage AUROC rises from 0.64 on observed time (A/B/C) to 0.97 on pseudotime (D/E/F) — reaching 1.00 in the joint Scenario F — so that PRESCIENT ranks first in every pseudotime scenario. MIOFlow gains more modestly on both datasets (0.66 $\rightarrow$ 0.72 on GSE230659, 0.67 $\rightarrow$ 0.76 on GSE178325), scNODE gains only on GSE178325 (0.60 $\rightarrow$ 0.69), and PRESCIENT is flat on GSE178325 (0.62 $\rightarrow$ 0.62). This reproduces scTimeBench’s observation on reprogramming data: pseudotime can denoise the temporal ordering and sharpen lineage recovery, but the improvement is not guaranteed and is largest where the observed-time sampling is least informative for a given method.

3.10 Embedding coherence is low in absolute terms

Absolute embedding ARI is low for all methods (≤ 0.23 ; Fig. 9), echoing scTimeBench’s finding that projected latent spaces often fail to resolve cell types cleanly. PRESCIENT is most coherent

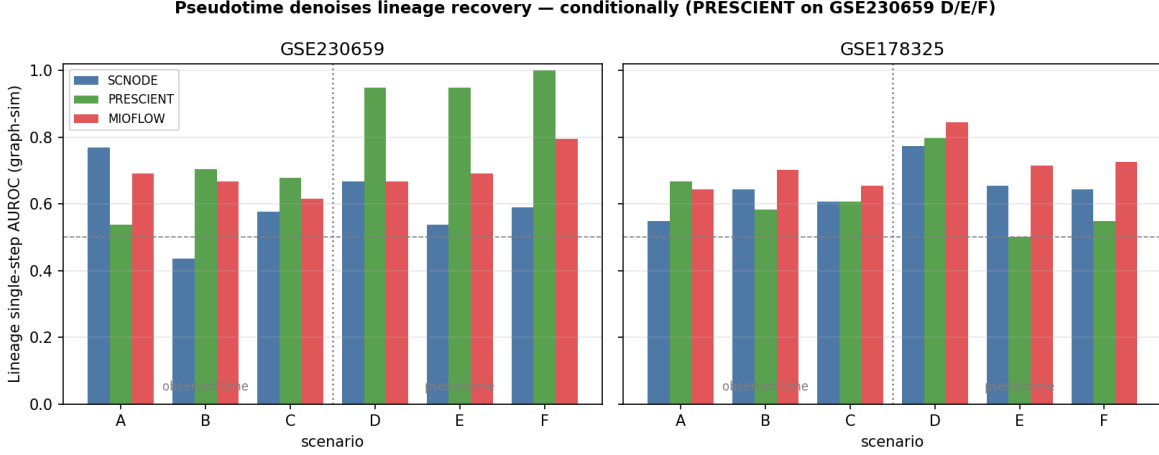


Figure 8. Pseudotime vs observed time. Single-step lineage AUROC (graph-sim) by method and scenario; A/B/C use observed time, D/E/F use a 15-bin diffusion-pseudotime axis (dotted divider). PRESCIENT’s lineage recovery on GSE230659 jumps under pseudotime (to 1.00 in Scenario F); gains elsewhere are milder and method-dependent.

on average, but the ordering changes by dataset and scenario, so embedding ARI is best read as a relative diagnostic against the frozen milestone labels.

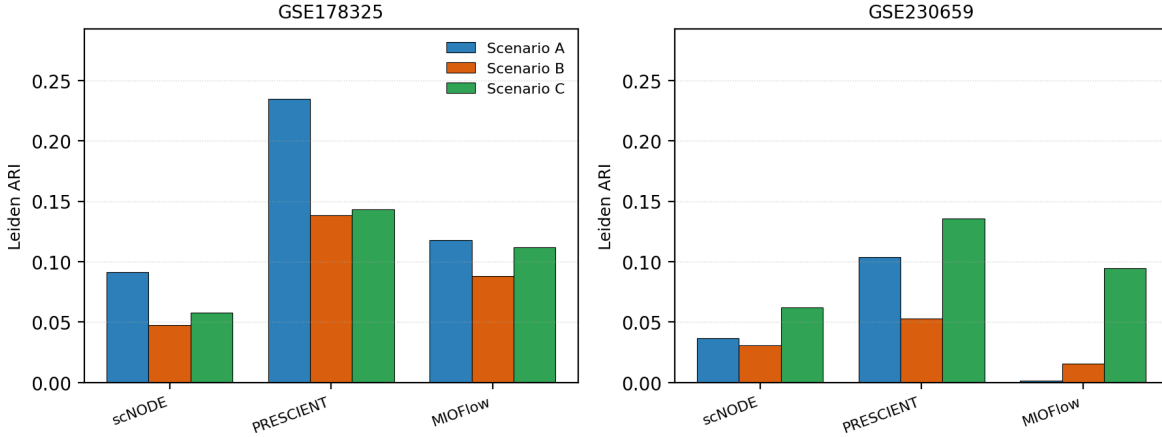


Figure 9. Embedding Coherence (Leiden-cluster ARI vs reference milestones) by method and scenario. Absolute ARI is low throughout and the method ordering is not stable across the two datasets.

4 Discussion

Narrowing a temporal-trajectory benchmark to the pluripotency domain turns the question from “which method wins on average?” into “does a method’s ranking transfer across the routes to pluripotency?”, and the answer here is: only for some tasks. PRESCIENT, MIOFlow, and scNODE trade the lead across forecast, embedding, and lineage tasks, and all three score substantially higher on OSKM than on chemical reprogramming. Crucially, the instability is task-dependent: lineage-fidelity rankings are near-discordant across the four marker-silver systems (Kendall’s $W = 0.19$),

reversing completely between the human hADSC and mouse systems, whereas forecast rankings are comparatively portable ($W = 0.75$). The non-transfer is therefore not a blanket caveat but a specific one — it is the lineage task, the task scTimeBench already flags as the field’s weakest, whose leaderboard is least trustworthy, and it breaks down along biological axes (species, starting cell type) rather than at random. Practically, a method should be chosen for the task that matters and validated on a system close to the target biology — a lineage-recovery result in particular should never be read off a single leaderboard. The reference-construction pipeline makes this domain benchmark feasible despite missing labels: scANVI/scArches transfer supplies cell-state references, and the geometric OOD gate — which catches a confident cross-modality collapse that classifier confidence entirely misses — decides when a transferred reference may be used. The two contributions reinforce each other: the cross-modality generalization result is the empirical reason the OOD gate refuses cross-modality references.

No single best method is the finding, not a gap. That winners differ by task — PRESCIENT for embedding coherence, MIOFlow or scNODE for lineage depending on the system, the forecasters splitting by scenario — is a positive result for practitioners, not a weakness of the study: method choice should be driven by the analytical objective rather than by a headline leaderboard, and a benchmark that reported only an averaged rank would hide exactly the information a user needs.

Practical guidance. For *forecasting*, prioritise the exact OT-loss metrics (Wasserstein/MMD) and treat scNODE/PRESCIENT as strong defaults, since forecast rankings are the most portable across systems ($W = 0.75$). For *lineage*, do not read recovery off embedding ARI — the two tasks rank methods differently — and interpret absolute AUROC against the empirical permutation floor (≈ 0.61 on these small graphs, Section 3.7), not against 0.5. For *reference transfer*, reject a transferred reference when the geometric OOD gate flags a large fraction of query cells, enriched over the reference’s own self-flag baseline (here cross-modality transfers, $6.5\text{--}9.7\times$); accept only within-modality, gate-passing transfers. And read any *single-seed* leaderboard as one draw: where 5-seed intervals exist the leader can change, so rank gaps inside a confidence interval should not be over-interpreted.

Scope of the claims. We frame this work as a *silver-reference benchmark and cautionary analysis*, not a biological-ground-truth study. Every lineage and embedding number is scored against marker-defined or scANVI-transferred references, so the results establish how method rankings behave — and how fragile they are — under a transparent, reproducible reference construction, rather than a definitive biological ordering of methods or a mechanism-discovery claim. “Recovery” throughout denotes agreement with these frozen coarse reference graphs, not recovery of ground-truth lineage. The three additional generative methods (scIMF, PI-SDE, Squidiff) are planned extensions whose model code is not vendored, raise an explicit `VendoringRequired` contract, and contribute to no result in this paper.

Limitations. Silver references are marker-defined or scANVI-transferred, not author-validated, so lineage fidelity measures alignment with a coarse reference graph rather than validated biological lineage; the reference graphs are small (3–4 edges for the marker-silver systems, 9 for the OSKM author-cluster protocol), so the lineage metrics are reported with this structural sparsity in mind and the OSKM system, built under a different reference protocol, is shown stratified rather than pooled with the marker-silver systems. Four of six datasets carry completed runs across all three tasks

(GSE242424 under the separate author-cluster protocol; GSE175634 carries forecast and embedding) while the organoid extension needs its own reference; the GSE218855/GSE298212 milestone markers are canonical rather than author-curated and so those two systems are framed as external stress tests rather than co-equal validation; and PRESCIENT’s formal runs were CPU-only. Multi-seed 95% confidence intervals are reported for the two primary datasets (Fig. 5); extending them to the cross-system datasets and the pseudotime scenarios is the main outstanding analysis, so the headline tables remain single-seed point estimates that — as the robustness analysis shows — should be read with the seed sensitivity in mind rather than as resolved rankings. The single-seed combined-rank gaps (e.g. 1.79 vs 1.85) are therefore descriptive, not statistically resolved. The three additional generative methods (scIMF, PI-SDE, Squidiff) are scoped as planned extensions whose model code is not yet vendored, and are excluded from all results.

5 Data and code availability

All datasets are public (NCBI GEO: GSE178325, GSE230659, GSE242424, GSE175634, GSE298212, GSE218855; ArrayExpress: E-MTAB-7552). The benchmark framework, capability-gated evaluators, frozen providers, scANVI/OOD utilities, Shirokane job scripts, and the scripts that regenerate every table and figure are in the project repository under `benchmark/` and `src/`. Each single-seed number reported here is read directly from a result file under `benchmark/reports/`, `benchmark/results/`, or `results/annotation_branch/`; the claim-to-source map is in `manuscript/REPRODUCIBILITY.md`. The multi-seed confidence intervals (Fig. 5) are a *derived* artifact summarised in `multiseed_ci.csv`. All 90 seed-runs (3 methods $\times$ 2 datasets $\times$ A/B/C $\times$ 5 seeds) executed on the HPC; the run logs (`logs/run_multiseed_benchmark.122804086.{1..90}.log`) and a SHA-256 checksum of every seed config (`multiseed_config_checksums.sha256`) are in the repository, and `multiseed_manifest.csv` maps each run to its config hash, output directory, and log. The raw per-seed metric JSONs are not synced (size) but are regenerated by the documented array recipe in `manuscript/REPRODUCIBILITY.md`. Nothing reported here is fabricated.

References

- Chen Y, Li S, *et al.* A fast chemical reprogramming system promotes cell identity transition through a diapause-like state. *Nature Cell Biology* 25:1146–1156 (2023). [doi:10.1038/s41556-023-01193-x](https://doi.org/10.1038/s41556-023-01193-x). (GSE218855.)
- Elorbany R, Popp JM, Rhodes K, *et al.* Single-cell sequencing reveals lineage-specific dynamic genetic regulation of gene expression during human cardiomyocyte differentiation. *PLoS Genetics* 18(1):e1009666 (2022). [doi:10.1371/journal.pgen.1009666](https://doi.org/10.1371/journal.pgen.1009666). (GSE175634.)
- Guan J, Wang G, Wang J, *et al.* Chemical reprogramming of human somatic cells to pluripotent stem cells. *Nature* 605:325–331 (2022). [doi:10.1038/s41586-022-04593-5](https://doi.org/10.1038/s41586-022-04593-5). (GSE178325.)
- He S, Zhu Y, *et al.* Squidiff: predicting cellular development and responses to perturbations using a diffusion model. *Nature Methods* 23(1):65–77 (2026).
- Huguet G, Magruder DS, Tong A, *et al.* Manifold interpolating optimal-transport flows for trajectory inference (MIOFlow). *Advances in Neural Information Processing Systems* 35:29705–29718 (2022). [arXiv:2206.14928](https://arxiv.org/abs/2206.14928).

- Jiang Q, Li L, *et al.* Learning collective multicellular dynamics with an interacting mean field neural SDE model (scIMF). *PLOS Computational Biology* 22(1):e1013916 (2026).
- Jiang Q, Wan L. A physics-informed neural SDE network for learning cellular dynamics from time-series scRNA-seq data (PI-SDE). *Bioinformatics* 40(Suppl. 2):ii120–ii127 (2024).
- Kanton S, Boyle MJ, He Z, *et al.* Organoid single-cell genomic atlas uncovers human-specific features of brain development. *Nature* 574:418–422 (2019). doi:10.1038/s41586-019-1654-9. (E-MTAB-7552.)
- Liuyang S, Wang G, Wang Y, *et al.* Highly efficient and rapid generation of human pluripotent stem cells by chemical reprogramming. *Cell Stem Cell* 30:450–459.e9 (2023). doi:10.1016/j.stem.2023.02.008. (GSE230659.)
- Lotfollahi M, Naghipourfar M, Luecken MD, *et al.* Mapping single-cell data to reference atlases by transfer learning (scArches). *Nature Biotechnology* 40:121–130 (2022). doi:10.1038/s41587-021-01001-7.
- Osakwe A, Huang EH, Li Y. scTimeBench: a streamlined benchmarking platform for single-cell time-series analysis. *bioRxiv* (2026). doi:10.64898/2026.03.16.712069.
- Peng F, *et al.* Chemical reprogramming of human blood cells to pluripotent stem cells. *Cell Stem Cell* 32:1192–1199.e11 (2025). doi:10.1016/j.stem.2025.07.003. (GSE298212.)
- Schiebinger G, Shu J, Tabaka M, *et al.* Optimal-transport analysis of single-cell gene expression identifies developmental trajectories in reprogramming (Waddington-OT). *Cell* 176:928–943.e22 (2019). doi:10.1016/j.cell.2019.01.006. (GSE115943.)
- Weiler P, Lange M, Klein M, Pe’er D, Theis FJ. CellRank 2: unified fate mapping in multiview single-cell data. *Nature Methods* 21:1196–1205 (2024). doi:10.1038/s41592-024-02303-9.
- Xu C, Lopez R, Mehlman E, Regier J, Jordan MI, Yosef N. Probabilistic harmonization and annotation of single-cell transcriptomics data with deep generative models (scANVI). *Molecular Systems Biology* 17:e9620 (2021). doi:10.15252/msb.20209620.
- Yeo GHT, Saksena SD, Gifford DK. Generative modeling of single-cell time series with PRESCIENT. *Nature Communications* 12:3222 (2021). doi:10.1038/s41467-021-23518-w.
- Zhang J, Larschan E, Bigness J, Singh R. scNODE: generative model for temporal single-cell transcriptomic data prediction. *Bioinformatics* 40(Suppl. 2):ii146–ii154 (2024). doi:10.1093/bioinformatics/btae393.

Table 5. Complete results matrix: every *completed* method on all six systems, observed-time scenarios A/B/C (means). Forecast = exact OT-loss Wasserstein (lower better; GSE230659 is on a different, unnormalised scale, cf. Fig. 2); Embed. = coherence ARI (higher better); Lineage = single-step graph-sim AUROC (higher better, read against the permutation floor ≈ 0.62 , Section 3.7). WOT/CellRank2 are lineage-only by construction; the three planned methods are not run. [†]GSE242424 (OSKM) uses the author-cluster reference protocol and is not pooled with the marker-silver systems.

Method	System	Forecast WD ↓	Embed. ARI ↑	Lineage AUROC ↑
scNODE	GSE178325	0.038	0.115	0.599
	GSE230659	204.8	0.067	0.594
	GSE218855	0.075	0.077	0.872
	GSE298212	0.045	0.000	0.735
	GSE175634	0.047	0.350	0.946
	GSE242424 [†]	0.035	0.147	0.795
PRESCIENT	GSE178325	0.042	0.080	0.619
	GSE230659	157.7	0.219	0.641
	GSE218855	0.075	0.070	0.778
	GSE298212	0.044	0.017	0.863
	GSE175634	0.042	0.184	0.523
	GSE242424 [†]	0.027	0.199	0.786
MIOFlow	GSE178325	0.046	0.096	0.667
	GSE230659	260.5	0.021	0.658
	GSE218855	0.076	0.117	0.761
	GSE298212	0.054	0.044	0.803
	GSE175634	0.038	0.366	0.970
	GSE242424 [†]	0.028	0.156	0.856
WOT	GSE178325	—	—	0.607
	GSE230659	—	—	0.513
	GSE218855	—	—	0.872
	GSE298212	—	—	0.795
	GSE175634	—	—	1.000
	GSE242424 [†]	—	—	0.939
CellRank2	GSE178325	—	—	0.583
	GSE230659	—	—	0.564
	GSE218855	—	—	0.923
	GSE298212	—	—	0.821
	GSE175634	—	—	1.000
	GSE242424 [†]	—	—	0.934
<i>scIMF, PI-SDE, Squidiff — planned extensions, not run (see Table 2).</i>				